Hypothesis Testing

Exercise

Jim Nasium claims that the mean A.C.T. score attained by two year college students is 22. Al Dente suspects this claim is too low and selects a random sample of 121 two year college students. The mean of the sample is 23.3 and the population standard deviation is assumed to be 3.3. Test at the 5% significance level. Show all steps.

Exercise

Solution:

Exercise2

Does the evidence support the idea that the average lecture consists of 3000 words if a random sample of the lectures of 16 professors had a mean of 3472 words, given the population standard deviation is 500 words? Use $\alpha = 0.01$. Assume that lecture lengths are approximately normally distributed. Show all steps.

Exercise2

Solution:

Exercise3

The Dog Days Lawn Service advertises that it will completely maintain your lawn at an average cost per customer of $35 per month. Assume the costs are normally distributed. A random sample of 18 Dog Days customers shows the average cost to be $32.50 with a population standard deviation of $8.10. Do the data support the claim that the average cost per month is less than $35.00? Use a 5% level of significance. Show all steps.

Exercise3

Solution:

Exercise4

Suppose that scores on the Scholastic Aptitude Test form a normal distribution with $\mu = 500$ and $\sigma = 100$. A high school counselor has developed a special course designed to boost SAT scores. A random sample of 16 students is selected to take the course and then the SAT. The sample had an average score of $\bar{x} = 544$. Does the course boost SAT scores? Test at $\alpha = 0.01$. Show all steps.

Exercise4

Solution:

Exercise5

- The secretary of an association of professional landscape gardeners claims that the average cost of services to customers is $90 per month. Feeling that this figure is too high, we question a random sample of 14 customers. Our sample yields a mean cost of $85 and a standard deviation of $10. Test at the 0.10 significance level. Assume that such costs are normally distributed.

Exercise5

Solution:

Exercise6

A study was done in Europe recently to investigate the health hazards of working long hours in front of a computer or word processor video displays. It found that it took an average of 2.6 hours before a certain symptom of eye strain developed. If a similar experiment in the United States using a sample of 85 people had a mean of 2.8 hours, with standard deviation of 1.25 hours, would this indicate that the American results are in conflict with the European results? Use a level of significance of 0.01.

Exercise6

Solution:

Exercise7

- A mean grade point average (GPA) of graduating college seniors who have been admitted to graduate school is 3.1, where an A is given 4 points. At Ivy University a random sample of 43 incoming graduate students yielded a GPA of 3.2 with a standard deviation of 0.42. Can we claim that the students going to Ivy have better grades than the national average, using the 0.05 significance level?

Exercise7

Solution:

Exercise8

Count and Countess Dracula supervise students who are training to be hematologists. For one project their 8 students had to count certain cell types in blood samples. Their counts were 103, 75, 82, 107, 63, 102, 81, and 72. Does this support the hypothesis that the mean count is 100? Use $\alpha = 0.05$. Assume that the cell counts are normally distributed and $\bar{x} = 85.63$ and $s = 16.35$.

Exercise8

Solution: